

Convexity

V. Leclère (ENPC)

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Why should I bother to learn this stuff?

- Convex vocabulary and results are needed throughout the course, especially to obtain optimality conditions and duality relations.
- Convex analysis tools like Fenchel transform appears in modern machine learning theory
- $\implies$ fundamental for M2 in continuous optimization
- $\implies$ useful for M2 in operation research, machine learning (and some part of probability or mechanics)

Contents

1 Convex sets [BV 2]

- Fundamental definitions
- Separation theorems

2 Convex functions [BV 3]

- definitions
- Convex function and optimization
- Some results on convex functions

3 Convex analysis

- Subdifferential
- Fenchel transform

4 Wrap-up

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Affine sets



Let $C \subset \mathbb{R}^n$

- C is **affine** if it contains the entire line through any two distinct points of C , i.e.,

$$\forall x, y \in C, \quad \forall \theta \in \mathbb{R}, \quad \theta x + (1 - \theta)y \in C.$$

- The **affine hull** of C is the set of **affine combination** of elements of C ,

$$\text{aff}(C) := \left\{ \sum_{i=1}^K \theta_i x_i \mid \forall x_i \in C, \forall \theta_i \in \mathbb{R}, \sum_{i=1}^K \theta_i = 1, \forall i \in [K], \forall K \in \mathbb{N} \right\}$$

- $\text{aff}(C)$ is the smallest affine space containing C .
- The **affine dimension** of C is the dimension of $\text{aff}(C)$ (i.e., the dimension of the vector space $\text{aff}(C) - x_0$ for $x_0 \in C$).
- The **relative interior** of C is defined as

$$\text{ri}(C) := \left\{ x \in C \mid \exists r > 0, \quad B(x, r) \cap \text{aff}(C) \subset C \right\}$$

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- The **relative interior** of C is defined as

$$\text{ri}(C) := \left\{ x \in C \mid \exists r > 0, \quad B(x, r) \cap \text{aff}(C) \subset C \right\}$$



- C is **convex** if for any two points x and y in C the segment $[x, y] \subset C$, i.e.,

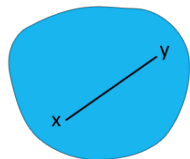
$$\forall x, y \in C, \forall \theta \in [0, 1], \theta x + (1 - \theta)y \in C.$$

- The **convex hull** of C is the set of **convex combination** of elements of C , i.e.,

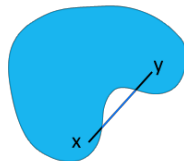
$$\text{conv}(C) := \left\{ \sum_{i=1}^K \theta_i x_i \mid \forall x_i \in C, \right. \\ \left. \forall \theta_i \in [0, 1], \sum_{i=1}^K \theta_i = 1, \forall i \in [K], \forall K \in \mathbb{N} \right\}$$

- $\text{conv}(C)$ is the smallest convex set containing C .

Convex set



Non - convex set



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- C is a **cone** if for all $x \in C$ the **ray** $\mathbb{R}_+x \subset C$, i.e.,

$$\forall x \in C, \quad \forall \theta \in \mathbb{R}_+, \quad \theta x \in C.$$

- The (convex) **conic hull** of C is the set of all **conic combination** of elements of C i.e.,

$$\text{cone}(C) := \left\{ \sum_{i=1}^K \theta_i x_i \mid \forall x_i \in C, \forall \theta_i \in \mathbb{R}_+, \forall i \in [K], \forall K \in \mathbb{N} \right\}$$

- $\text{cone}(C)$ is the smallest **convex** cone containing C .
- A cone C is **pointed** if it does not contain any full line $\mathbb{R}x$ for $x \neq 0$.
- For C convex, $\text{cone}(C) = \bigcup_{t \geq 0} tC$

Examples



- If C is convex then, for $x \in C$,

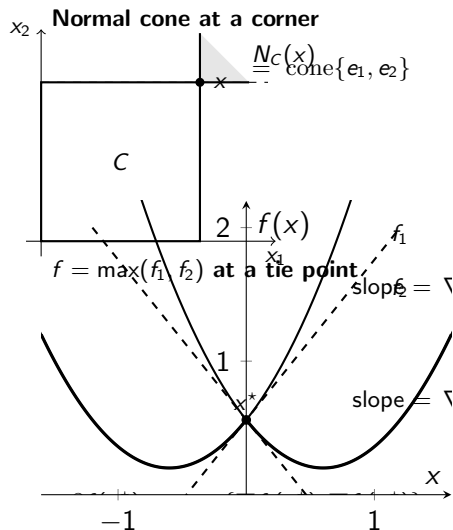
$$\partial(\mathbb{I}_C)(x) = N_C(x).$$

♣ Exercise: Prove it.

- If f_1 and f_2 are convex and differentiable.

Define $f = \max(f_1, f_2)$. Then

- ▶ if $f_1(x) > f_2(x)$, $\partial f(x) = \{\nabla f_1(x)\}$
- ▶ if $f_1(x) < f_2(x)$, $\partial f(x) = \{\nabla f_2(x)\}$
- ▶ if $f_1(x) = f_2(x)$,
 $\partial f(x) = \overline{\text{conv}}(\{\nabla f_1(x), \nabla f_2(x)\})$.





Assume that all sets denoted by C (indexed or not) are convex.

- $C_1 + C_2$ and $C_1 \times C_2$ are convex sets.
- For any arbitrary index set $\mathcal{I}$ the intersection $\bigcap_{i \in \mathcal{I}} C_i$ is convex.
- Let f be an affine function. Then $f(C)$ and $f^{-1}(C)$ are convex.
- In particular, $C + x_0$, and tC are convex. The projection of C on any affine space is convex.
- The closure $\text{cl}(C)$ and relative interior $\text{ri}(C)$ are convex.

♣ Exercise: Prove these results.

Perspective and linear-fractional function



Let $P : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ be the **perspective function** defined as $P(x, t) = x/t$, with $\text{dom}(P) = \mathbb{R}^n \times \mathbb{R}_+^*$.

Theorem

If $C \subset \text{dom}(P)$ is convex, then $P(C)$ is convex.

If $C \subset \mathbb{R}^n$ is convex, then $P^{-1}(C)$ is convex.

♠ Exercise: Prove this result.

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If $C \subset \mathbb{R}^n$ is convex, then $P^{-1}(C)$ is convex.

♠ Exercise: Prove this result.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a **linear-fractional function** of the form $f(x) := (Ax + b)/(c^\top x + d)$, with $\text{dom}(f) = \{x \mid c^\top x + d > 0\}$.

Theorem

If $C \subset \text{dom}(f)$ is convex, then $f(C)$ and $f^{-1}(C)$ are convex.

♣ Exercise: prove this result.

Cone ordering

Let $K \subset \mathbb{R}^n$ be a closed, convex, pointed cone with non-empty interior. We define the **cone ordering** according to K by

$$x \preceq_K y \iff y - x \in K.$$

Examples:

- For $K = \mathbb{R}_+^n$, $\preceq_K$ is the **componentwise ordering**, i.e.,

$$x \preceq_{\mathbb{R}_+^n} y \iff \forall i \in [n], x_i \leq y_i.$$

- For $K = S_n^+(\mathbb{R})$, $\preceq_K$ is the **Löwner ordering**, i.e.,

$$A \preceq_{S_n^+} B \iff B - A \in S_n^+(\mathbb{R}).$$

♣ Exercise: Prove that $\preceq_K$ is a partial order (i.e., reflexive, antisymmetric, transitive) compatible with scalar product, addition and limits.

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Separation



Let X be a Banach space, and X^* its **topological dual** (i.e. the set of all continuous linear forms on X).

Theorem (Simple separation)

Let A and B be convex non-empty, disjoint subsets of X . Assume that A is open (or B is open). There exists a **separating hyperplane** $(x^*, \alpha) \in X^* \times \mathbb{R}$ such that

$$\langle x^*, a \rangle < \alpha \leq \langle x^*, b \rangle \quad \forall a, b \in A \times B.$$

Theorem (Strong separation)

Let A and B be convex non-empty, disjoint subsets of X . Assume that, A is closed, and B is compact (e.g. a point), then there exists a **strict separating hyperplane** $(x^*, \alpha) \in X^* \times \mathbb{R}$ such that, there exists $\varepsilon > 0$,

$$\langle x^*, a \rangle + \varepsilon \leq \alpha \leq \langle x^*, b \rangle - \varepsilon \quad \forall a, b \in A \times B.$$

Remark: In Banach spaces these theorems require the Zorn Lemma which is equivalent to the axiom of choice. We will stay in finite dimension where these theorems are easier to prove.



Theorem

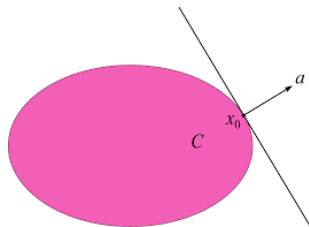
Let $C \subset \mathbb{R}^n$ be a non-empty convex set and $x_0 \in C \setminus \text{ri}(C)$. Then there exists $a \neq 0$ such that

$$a^\top x \geq a^\top x_0, \quad \forall x \in C$$

Then $H = \{x \mid a^\top x = a^\top x_0\}$ is a **supporting hyperplane** of C at x_0 .

♣ Exercise: prove this theorem

Remark: there can be more than one supporting hyperplane at a given point.





- The **closed convex hull** of $C \subset X$, denoted $\overline{\text{conv}}(C)$ is the smallest closed convex set containing C .
- $\overline{\text{conv}}(C)$ is the intersection of all the **closed** half-spaces containing C .
- A polyhedron is a finite intersection of half-spaces while a **closed** convex set is a possibly infinite intersection of closed half-spaces.

Dual and normal cones

- Let $C \subset \mathbb{R}^n$ be a set. We define its (positive) **dual cone** by

$$C^\oplus := \{x \mid x^\top c \geq 0, \quad \forall c \in C\}$$

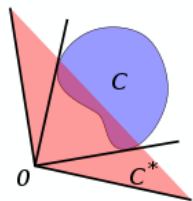
- For any set C , $C^\oplus$ is a closed convex cone.
- The **normal cone** of C at x_0 is

$$N_C(x_0) := \{\lambda \in \mathbb{R}^n \mid \lambda^\top (x - x_0) \leq 0, \forall x \in C\}$$

(and $N_C(x_0) = \emptyset$ if $x_0 \notin C$).

Let $K \subset \mathbb{R}^n$ be a cone.

- $K^\oplus$ is closed convex.
- $K_1 \subset K_2$ implies $K_2^\oplus \subset K_1^\oplus$
- $K^{\oplus\oplus} = \overline{\text{conv}} K$



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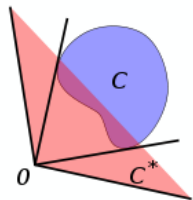
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Examples

- The positive orthant $K = \mathbb{R}_+^n$ is a **self dual** cone, that is $K^\oplus = K$.
- In the space of symmetric matrices $S_n(\mathbb{R})$, with the scalar product $\langle A, B \rangle = \text{tr}(AB)$, the set of positive semidefinite matrices $K = S_n^+(\mathbb{R})$ is self dual.
- Let $\|\cdot\|$ be a norm. The cone $K = \{(x, t) \mid \|x\| \leq t\}$ has for dual $K^\oplus = \{(\lambda, z) \mid \|\lambda\|_\star \leq z\}$, where $\|\lambda\|_\star := \sup_{x: \|x\| \leq 1} \lambda^\top x$.

♠ Exercise: prove these results

Video resources

https://www.youtube.com/watch?v=P3W_wFZ2kUo

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Functions with non finite values



- It is very useful in optimization to allow functions to take non-finite values, that is to take values in $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$.
- If both $-\infty$ and $+\infty$ are allowed be very careful of each addition¹!
- Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$. We define
 - ▶ The **epigraph** of f as

$$\text{epi}(f) := \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}$$

- ▶ the **domain** of f as

$$\text{dom}(f) := \{x \in \mathbb{R}^n \mid f(x) < +\infty\}.$$

- ▶ The **sublevel set** of level α

$$\text{lev}_\alpha(f) := \{x \in \mathbb{R}^n \mid f(x) \leq \alpha\}.$$

- f is said to be **lower semicontinuous** (l.s.c.) if $\text{epi}(f)$ is closed.
- f is said to be **proper** if it never takes value $-\infty$, has a non-empty domain (at least one finite value).

¹Which is why we usually consider proper functions.



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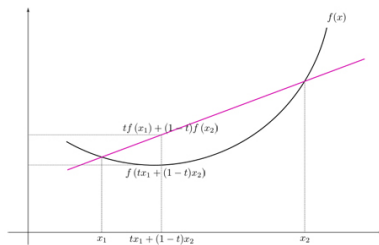
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- A function $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is **convex** if its epigraph is convex.
- $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex iff

$$\forall t \in [0, 1], \forall x, y \in \mathbb{R}^n, \\ f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

- f is **concave** if $-f$ is convex.





- If f, g convex, $t > 0$, then $tf + g$ is convex.
- If f convex non-decreasing, g convex, then $f \circ g$ convex.
- If f convex and a affine, then $f \circ a$ is convex.
- If $(f_i)_{i \in I}$ is a family of convex functions, then $\sup_{i \in I} f_i$ is convex.
- The domain and the sublevel sets of a convex function are convex.
- A convex function is always above its tangents (supporting hyperplanes).

♣ Exercise: Prove these results.



- If f, g convex, $t > 0$, then $tf + g$ is convex.
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Theorem (Jensen inequality)

Let f be a convex function and $\mathbf{X}$ an integrable random variable. Then we have

$$f(\mathbb{E}[\mathbf{X}]) \leq \mathbb{E}[f(\mathbf{X})].$$



Consider a convex function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$.

- f is continuous (on $\mathbb{R}^n$) if and only if $\text{dom}(f) = \mathbb{R}^n$ (i.e., if it is finite everywhere).
- f is continuous on the interior of its domain.
- f is Lipschitz on any compact convex set contained in the interior of its domain.

Convex functions: strict and strong convexity



- $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is strictly convex iff

$$\forall t \in]0, 1[, \quad \forall x, y \in \mathbb{R}^n, \quad f(tx + (1 - t)y) < tf(x) + (1 - t)f(y)$$

- $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is α -strongly convex iff $\forall t \in]0, 1[, \quad \forall x, y \in \mathbb{R}^n,$

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) - \frac{1}{2}\alpha t(1 - t)\|x - y\|^2$$

- If $f \in C^1(\mathbb{R}^n)$

- ▶ $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$ iff f convex
- ▶ if strict inequality holds, then f strictly convex
- ▶ $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is α -strongly convex iff $\forall x, y \in \mathbb{R}^n$

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2}\|y - x\|^2$$

- If $f \in C^2(\mathbb{R}^n),$

- ▶ $\nabla^2 f \succcurlyeq 0$ iff f convex
- ▶ if $\nabla^2 f \succ 0$ then f strictly convex
- ▶ if $\nabla^2 f \succcurlyeq \alpha I$ then f is α -strongly convex

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- ▶ if $\nabla^2 f \succ 0$ then f strictly convex
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Three convexity tests you will actually use



Test 1 — Definition (segment / Jensen)

To prove f convex on a convex domain $\text{dom}(f)$, show

$$\forall \mathbf{x}, \mathbf{y} \in \text{dom}(f), \forall t \in [0, 1], \quad f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y}).$$

Test 2 — Epigraph / sublevel sets (geometric)

f is convex $\iff$ its epigraph is convex:

$$\text{epi}(f) = \{(x, s) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq s\} \text{ is convex.}$$

Test 3 — Smooth test (first/second order)

If $f \in C^1$ on an open convex set:

$$f \text{ convex} \iff \langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0 \quad \forall \mathbf{x}, \mathbf{y}.$$

If $f \in C^2$:

$$f \text{ convex} \iff \nabla^2 f(\mathbf{x}) \succeq 0 \quad \forall \mathbf{x}, \quad f \text{ } \alpha\text{-strongly convex} \iff \nabla^2 f(\mathbf{x}) \succeq \alpha I.$$

Important examples

- The **indicator function** of a set $C \subset \mathbb{R}^n$,

$$\mathbb{I}_C(x) := \begin{cases} 0 & \text{if } x \in C \\ +\infty & \text{otherwise} \end{cases}$$

is convex iff C is convex.

- $x \mapsto e^{ax}$ is convex for any $a \in \mathbb{R}$
- $x \mapsto \|x\|^q$ is convex for $q \geq 1$ and any norm
- $x \mapsto \ln(x)$ is concave
- $x \mapsto x \ln(x)$ is convex
- $x \mapsto \ln(\sum_{i=1}^n e^{x_i})$ is convex

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$$\underset{x \in C}{\text{Min}} \quad f(x)$$

Where C is non-empty, closed convex and f convex finite valued², is a **convex optimization problem**.

- If C is compact and f proper lsc, then there exists an optimal solution.
- If f is proper lsc and coercive, then there exists an optimal solution.
- The set of optimal solutions is convex.
- If f is strictly convex the minimum (if it exists) is unique.
- If f is α -strongly convex (and proper lsc) the minimum exists and is unique.

♣ Exercise: Prove these results.

²Alternatively, f can be proper lower semicontinuous. Here I assumed $C \subset \text{dom}(f)$.



Note that minimizing f over C or minimizing $f + \mathbb{I}_C$ over $\mathbb{R}^n$ is the same thing.

We consider the (unconstrained) optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{Min}} \quad f(x),$$

with $x^\sharp$ an optimal solution and f not necessarily convex.

- If f is differentiable, then $\nabla f(x^\sharp) = 0$.
- If f is twice differentiable, then $\nabla^2 f(x^\sharp) \succeq 0$.
- If f is twice differentiable and $\nabla f(x_0) = 0$ and $\nabla^2 f(x_0) \succ 0$ then x_0 is a local minimum.

If, in addition, f is convex then $\nabla f(x) = 0$ is a necessary and sufficient optimality condition.



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Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \bar{\mathbb{R}}$ be a jointly convex function and $C \subset \mathbb{R}^m$ be a convex set. The function

$$g : x \mapsto \inf_{y \in C} f(x, y)$$

is convex.

♠ Exercise: Prove this result.

♣ Exercise: Prove that the function **distance** to a convex set C defined by

$$d_C(x) := \inf_{c \in C} \|c - x\|$$

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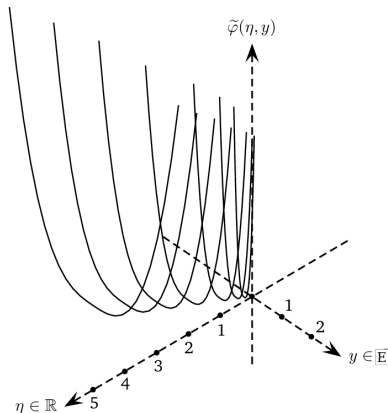
Let $\phi : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. The **perspective** of ϕ is defined as $\tilde{\phi} : \mathbb{R}_+^* \times \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$\tilde{\phi}(\eta, y) := \eta \phi(y/\eta).$$

Theorem

ϕ is convex iff $\tilde{\phi}$ is convex.

♠ Exercise: prove this result





Let f and g be proper function from $\mathbb{R}^n$ to $\overline{\mathbb{R}}$. We define

$$f \square g : x \mapsto \inf_{y \in \mathbb{R}^n} f(y) + g(x - y)$$

♣ Exercise: Show that

- $f \square g = g \square f$
- If f and g are convex then so is $f \square g$

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Subdifferential of convex function



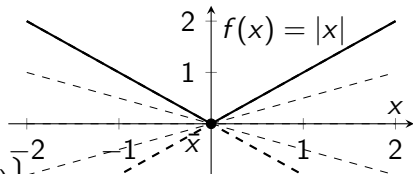
Let X be an Hilbert space, $f : X \rightarrow \bar{\mathbb{R}}$ convex.

- The **subdifferential** of f at $x \in \text{dom}(f)$ is the set of slopes of all affine minorants of f exact at x :

$$\partial f(x) := \left\{ \lambda \in X \mid \forall z \in X, f(z) \geq \langle \lambda, z - x \rangle + f(x) \right\}.$$

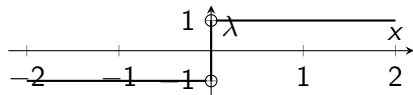
- If f is differentiable at x then

$$\partial f(x) = \{\nabla f(x)\}.$$



Supporting affine minorants of $f(x) = |x|$.

$\lambda \in \partial f(0)$ iff the line $y = \lambda x$ is a supporting line of f at 0.



Graph of $x \mapsto \partial f(x)$.

Examples



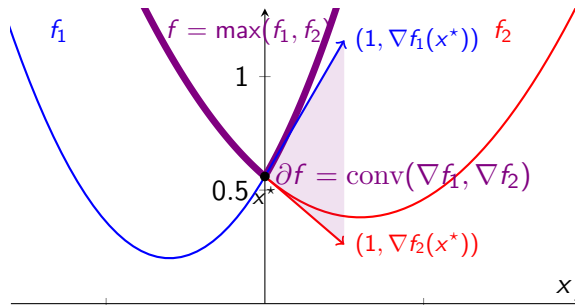
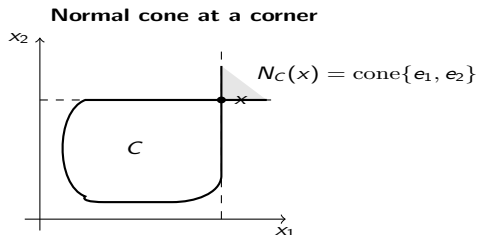
- If C is convex then, for $x \in C$,

$$\partial(\mathbb{I}_C)(x) = N_C(x).$$

♣ Exercise: Prove it.

- If f_1 and f_2 are convex and differentiable. Define $f = \max(f_1, f_2)$. Then

- ▶ if $f_1(x) > f_2(x)$,
 $\partial f(x) = \{\nabla f_1(x)\}$
- ▶ if $f_1(x) < f_2(x)$,
 $\partial f(x) = \{\nabla f_2(x)\}$
- ▶ if $f_1(x) = f_2(x)$, $\partial f(x) = \overline{\text{conv}}(\{\nabla f_1(x), \nabla f_2(x)\})$.



Subdifferential calculus



Let f_1 and f_2 be proper convex lsc functions.

Theorem (Moreau-Rockafellar)

We have

$$\partial(f_1)(x) + \partial(f_2)(x) \subset \partial(f_1 + f_2)(x), \quad \forall x$$

Further if $\text{ri}(\text{dom}(f_1)) \cap \text{ri}(\text{dom}(f_2)) \neq \emptyset$ then

$$\partial(f_1)(x) + \partial(f_2)(x) = \partial(f_1 + f_2)(x), \quad \forall x$$

When f_i is polyhedral you can replace $\text{ri}(\text{dom}(f_i))$ by $\text{dom}(f_i)$ in the condition.

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When f_i is polyhedral you can replace $\text{ri}(\text{dom}(f_i))$ by $\text{dom}(f_i)$ in the condition.

Theorem

If f is convex and $a : x \mapsto Ax + b$ with $\text{Im}(a) \cap \text{ri}(\text{dom}(f)) \neq \emptyset$, then

$$\partial(f \circ a)(x) = A^\top \partial f(Ax + b).$$

Counterexample to the qualification in Moreau–Rockafellar



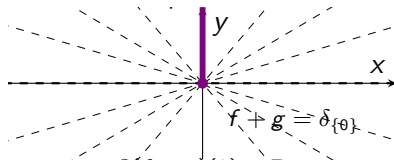
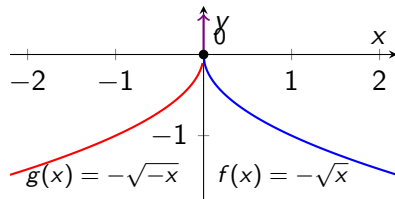
$$f(x) = \begin{cases} -\sqrt{x}, & x \geq 0, \\ +\infty, & x < 0, \end{cases} \quad g(x) = f(-x).$$

Then $\text{dom } f = [0, \infty)$, $\text{dom } g = (-\infty, 0]$, hence $\text{dom } f \cap \text{dom } g = \{0\}$ and

$$(f + g)(x) = \begin{cases} 0, & x = 0, \\ +\infty, & x \neq 0, \end{cases} \Rightarrow f + g = \delta_{\{0\}}.$$

Therefore $\partial(f + g)(0) = \mathbb{R}$. On the other hand, $\partial f(0) = \partial g(0) = \emptyset$, so

$$\partial f(0) + \partial g(0) = \emptyset \subsetneq \mathbb{R} = \partial(f + g)(0).$$





Theorem

Let $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper convex function (not necessarily differentiable). Then $x^\# \in \text{dom}(f)$ is a minimizer of f if and only if $0 \in \partial f(x^\#)$.



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Theorem

Let f be a proper convex function and C a closed non-empty convex set such that $\text{ri}(C) \cap \text{ri}(\text{dom}(f)) \neq \emptyset$ then $x^\#$ is an optimal solution to

$$\min_{x \in C} f(x)$$

iff

$$0 \in \partial f(x^\#) + N_C(x^\#),$$

iff

$$\exists \lambda \in \partial f(x^\#), \quad \lambda \in -N_C(x^\#).$$

Normal cone, Tangent cone and optimality

Let C be a convex set. We define the **tangent cone** of $C \subset \mathbb{R}^n$ at point $x \in C$, as the set of directions in which you can move from x while staying in C for some time, that is

$$T_C(x) := \overline{\left\{ \lambda(y - x) \mid y \in C, \lambda \in \mathbb{R}^+ \right\}}$$

In particular, $T_C(x) = \mathbb{R}^n$ iff $x \in \text{int}(C)$.

♣ Exercise: Prove that $[T_C(x)]^\oplus = -N_C(x)$.

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Partial infimum



Let $f : X \times Y \rightarrow \bar{\mathbb{R}}$ be a jointly convex and proper function, and define

$$v(x) = \inf_{y \in Y} f(x, y)$$

then v is convex.

If v is proper, and $v(x) = f(x, y^\sharp(x))$ then

$$\partial v(x) = \{g \in X \mid (g, 0) \in \partial f(x, y^\sharp(x))\}$$

proof:

$$\begin{aligned} g \in \partial v(x) &\Leftrightarrow \forall x', \quad v(x') \geq v(x) + \langle g, x' - x \rangle \\ &\Leftrightarrow \forall x', y' \quad f(x', y') \geq f(x, y^\sharp(x)) + \left\langle \begin{pmatrix} g \\ 0 \end{pmatrix}, \begin{pmatrix} x' \\ y' \end{pmatrix} - \begin{pmatrix} x \\ y^\sharp(x) \end{pmatrix} \right\rangle \\ &\Leftrightarrow \begin{pmatrix} g \\ 0 \end{pmatrix} \in \partial f(x, y^\sharp(x)) \end{aligned}$$

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- Assume f convex, then f is continuous on the relative interior of its domain, and Lipschitz on any compact contained in the relative interior of its domain.
- A proper convex function is subdifferentiable on the relative interior of its domain.
- If f is convex and finite everywhere on X (i.e. $\text{dom}(f) = X$), it is L -Lipschitz iff $\partial f(x) \subset B(0, L), \quad \forall x \in X$.

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Fenchel transform



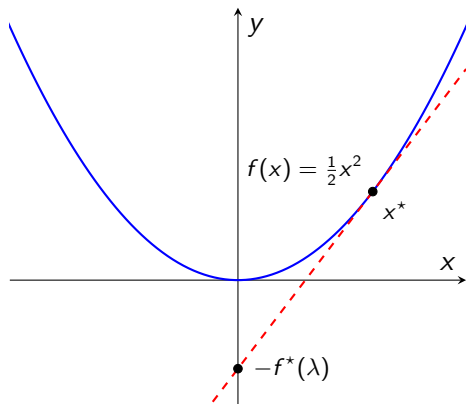
Let X be a Hilbert space, $f : X \rightarrow \bar{\mathbb{R}}$ be a proper function.

- The Fenchel transform of f , is $f^* : X \rightarrow \bar{\mathbb{R}}$ with

$$f^*(\lambda) := \sup_{x \in X} \langle \lambda, x \rangle - f(x).$$

- f^* is convex lsc as the supremum of affine functions.
- $f \leq g$ implies that $f^* \geq g^*$.
- If f is proper convex lsc, then $f^{**} = f$, otherwise $f^{**} \leq f$.

♣ Exercise: Prove the first two points



Fenchel transform and subdifferential



- By definition $f^*(\lambda) \geq \langle \lambda, x \rangle - f(x)$ for all x ,
- thus we always have (Fenchel-Young) $f(x) + f^*(\lambda) \geq \langle \lambda, x \rangle$.
- Recall that $\lambda \in \partial f(x)$ iff,

$$f(x') \geq f(x) + \langle \lambda, x' - x \rangle, \quad \forall x'$$

iff

$$\langle \lambda, x \rangle - f(x) \geq \langle \lambda, x' \rangle - f(x') \quad \forall x'$$

that is

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- If f proper convex lsc

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What you have to know

- What is a **affine set**, a **convex set**, a **polyhedron**, a (convex) **cone**
- What is a **convex** function, that it is above its tangents.
- Jensen inequality
- What is a convex optimization problem. That any local minimum is a global minimum.
- For $\min_{x \in C} f(x)$ with f differentiable, the necessary optimality condition $\nabla f(x^\#) \in [T_C(x^\#)]^\oplus$

What you really should know

- That you can separate convex sets with a linear function
- What is the positive dual of a cone
- Basic manipulations preserving convexity (sum, cartesian product, intersection, linear projection)
- What is the domain, the sublevel of a function f
- What is a lower semicontinuous function, a proper convex function
- Conditions of (strict, strong) convexity for differentiable functions
- The partial minimum of a convex function is convex
- The definition of the subdifferential.
- The definition of the Fenchel transform.
- The link between Fenchel transform and subdifferential.

What you have to be able to do

- Show that a set is convex
- Show that a function is (strictly, strongly) convex
- Go from constrained problem to unconstrained problem using the indicator function $\mathbb{I}_X$

What you should be able to do

- Compute dual cones
- Use advanced results (projection, partial infimum, perspective) to show that a function or a set is convex
- Compute the Fenchel transform of simple functions